Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 1 of 8 · Question 8 part a.ii · 2 marks

2024 JUN · Paper 01 · find part a.ii in this mark scheme extract

8 (a) (i)	• large waste/named waste/objects; • filtered/removed/trapped/separated;	reject refs to molecules	(2)
(ii)	• bacteria/microorganisms present; • require oxygen for aerobic respiration;		(2)
(iii)	methane;		(1)
(b)	• contains nitrates/nitrogen; • for plant growth		(2)

Total for Question 8 = 7 Marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 2 of 8 · Question 8 part a.i · 2 marks

2024 JUN · Paper 01 · find part a.i in this mark scheme extract

8 (a) (i)	• large waste/named waste/objects; • filtered/removed/trapped/separated;	reject refs to molecules	(2)
(ii)	• bacteria/microorganisms present; • require oxygen for aerobic respiration;		(2)
(iii)	methane;		(1)
(b)	• contains nitrates/nitrogen; • for plant growth		(2)

Total for Question 8 = 7 Marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 3 of 8 · Question 8 part a.iii · 1 mark

2024 JUN · Paper 01 · find part a.iii in this mark scheme extract

8 (a)	(i)	• large waste/named waste/objects; • filtered/removed/trapped/separated;	reject refs to molecules	(2)
	(ii)	• bacteria/microorganisms present; • require oxygen for aerobic respiration;		(2)
	(iii)	methane;		(1)
(b)		• contains nitrates/nitrogen; • for plant growth		(2)

Total for Question 8 = 7 Marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 4 of 8 · Question 8 part b · 2 marks

2024 JUN · Paper 01 · find part b in this mark scheme extract

8 (a) (i)	• large waste/named waste/objects; • filtered/removed/trapped/separated;	reject refs to molecules	(2)
(ii)	• bacteria/microorganisms present; • require oxygen for aerobic respiration;		(2)
(iii)	methane;		(1)
(b)	• contains nitrates/nitrogen; • for plant growth		(2)

Total for Question 8 = 7 Marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 5 of 8 · Question 5 part b · 2 marks

2024 JUN · Paper 02 · find part b in this mark scheme extract

Number				
5	(a) (i)	**C** the type of antibiotic added The following are incorrect because: A the mass of bacteria added is irrelevant B the size of the inhibition zone is the dependent variable D the mass of agar in the Petri dish is irrelevant		(1)
	(ii)	Any two from: • sterilise equipment/named equipment/disinfect/swab desk; • use of sterile agar; • reference to use of lid on Petri dish/ minimising the time the lid is removed; • Flame from Bunsen burner:		(2)
	(iii)	(antibiotic) Q; greater inhibition zone /greater clear zone/ more bacteria killed;		(2)
	(iv)	Any four from: • mutation; • (resistant) bacteria survive; • reproduce; • pass on resistant gene/allele; • Increase of resistant gene/allele in the population/ increase in allele frequency (over many generations)		(4)
	(b)	• (bacteria/fungi/decomposers) release enzymes/named enzyme • breakdown dead/organic matter/ named organic matter; • release nutrients/named nutrients into the soil/environment;	Ignore carbon dioxide Ignore fertiliser	(2)

Total for Question 5 = 11 marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 6 of 8 · Question 5 part a.i · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Question number					
5	(a)	(i)	• prevent contamination/no mixing with water; • because rainwater will flow away from well;		2
		(ii)	allow urine/liquid waste, to soak away/absorbed by soil;		1
		(iii)	• prevent smell; • covered to prevent entry of flies/rats; • as these carry bacteria/pathogens; • and spread disease; • long handle to keep hands away from pit entrance/cover faeces; • more likely to be bacteria/pathogens in that area;		5
	(b)		• any seven from • equal volumes of two solutions; • same concentrations; • wash concrete slab with one solution; • leave for specified time; • use sterile transfer loop or equivalent; • transfer from slab to Petri dish/nutrient agar; • incubate for specified time; • count colonies/measure inhibition zones; • repeat with second solution; • compare results;		7

Total 15

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 6 of 8 · Question 5 part a.i · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 7 of 8 · Question 5 part a.ii · 1 mark

2023 JUN · Paper 01 · extract 1 of 2 · find part a.ii in this mark scheme extract

Number				
5	(a) (i)	• prevent contamination/no mixing with water; • because rainwater will flow away from well;		2
	(ii)	allow urine/liquid waste, to soak away/absorbed by soil;		1
	(iii)	• prevent smell; • covered to prevent entry of flies/rats; • as these carry bacteria/pathogens; • and spread disease; • long handle to keep hands away from pit entrance/cover faeces; • more likely to be bacteria/pathogens in that area;		5
	(b)	• any seven from • equal volumes of two solutions; • same concentrations; • wash concrete slab with one solution; • leave for specified time; • use sterile transfer loop or equivalent; • transfer from slab to Petri dish/nutrient agar; • incubate for specified time; • count colonies/measure inhibition zones; • repeat with second solution; • compare results;		7

Total 15

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 7 of 8 · Question 5 part a.ii · 1 mark

2023 JUN · Paper 01 · extract 2 of 2 · find part a.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 8 of 8 · Question 5 part a.iii · 5 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Question number					
5	(a)	(i)	• prevent contamination/no mixing with water; • because rainwater will flow away from well;		2
		(ii)	allow urine/liquid waste, to soak away/absorbed by soil;		1
		(iii)	• prevent smell; • covered to prevent entry of flies/rats; • as these carry bacteria/pathogens; • and spread disease; • long handle to keep hands away from pit entrance/cover faeces; • more likely to be bacteria/pathogens in that area;		5
	(b)		• any seven from • equal volumes of two solutions; • same concentrations; • wash concrete slab with one solution; • leave for specified time; • use sterile transfer loop or equivalent; • transfer from slab to Petri dish/nutrient agar; • incubate for specified time; • count colonies/measure inhibition zones; • repeat with second solution; • compare results;		7

Total 15

Type MS · Explain sewage treatment, pit latrine design and decomposer roles

Answer 8 of 8 · Question 5 part a.iii · 5 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks